Analysis of Partial-Share (PS) Algorithm

Derivation Note

Algorithm 1 Adaptive Partial-Share (PS) Algorithm (Shared branch with IPS update)

Input: Tree $\mathcal{T}$, shared leaf set $\mathcal{L}_{\text{shr}}$, learning rate η, exploration base rate ϵ.
Notation: For a node i, let $\mathcal{L}(i)$ denote the set of leaf descendants of i.
Pre-processing: Label a non-leaf node i as *Safe* if $\mathcal{L}(i) \subseteq \mathcal{L}_{\text{shr}}$, and *Risky* otherwise.
Initialize:
  $\theta_\ell \leftarrow 0,\ \forall \ell \in \mathcal{L}$
  For every edge (i, j) with $j \in Children(i)$:

$$S_{\text{shr}}[i,j] \leftarrow \sum_{\ell \in \mathcal{L}(j) \cap \mathcal{L}_{\text{shr}}} \exp(\eta\theta_\ell), \qquad S_{\text{unshr}}[i,j] \leftarrow \mathbf{1}\{\mathcal{L}(j) \cap \mathcal{L}_{\text{unshr}} \neq \emptyset\}.$$

  Interpret $S_{\text{unshr}}[i,j]$ as a maintained unshared aggregate state, updated online along sampled unshared paths.
for each round $t = 1, \ldots, T$ **do**
  1. Adaptive Selection
  $i \leftarrow root$, $\Pi_t(root) \leftarrow 1$
  Initialize sampled path $\mathcal{P}_t \leftarrow \emptyset$
  while i is not a leaf **do**
    for each $j \in Children(i)$ **do**

$$P_t^{\exp}(j \mid i) \leftarrow \frac{S_{\text{shr}}[i,j] + S_{\text{unshr}}[i,j]}{\sum_{k \in Children(i)} \big(S_{\text{shr}}[i,k] + S_{\text{unshr}}[i,k]\big)}$$

    end for

$$\epsilon_i \leftarrow \epsilon \cdot \mathbf{1}\{i \text{ is Risky and } Children(i) \not\subseteq \mathcal{L}\}$$

    for each $j \in Children(i)$ **do**

$$p_t(j \mid i) \leftarrow (1 - \epsilon_i) P_t^{\exp}(j \mid i) + \frac{\epsilon_i}{|Children(i)|}$$

    end for
    Sample $j_t \sim p_t(\cdot \mid i)$
    Append (i, j_t) to $\mathcal{P}_t$
    $\Pi_t(j_t) \leftarrow \Pi_t(i)\, p_t(j_t \mid i)$
    $i \leftarrow j_t$
  end while
  Let sampled leaf be $\ell_t \leftarrow i$. Observe terminal cost c_t.
  2. Hybrid Update
  if $\ell_t \in \mathcal{L}_{\text{shr}}$ **then** ▷ **Shared branch**
    $val_{\text{old}} \leftarrow \exp(\eta\theta_{\ell_t})$
    $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - \frac{c_t}{\Pi_t(\ell_t)}$
    $\Delta_t \leftarrow \exp(\eta\theta_{\ell_t}) - val_{\text{old}}$
    Back-propagation: Propagate Δ_t to S_{shr} along sampled ancestors:

$$S_{\text{shr}}[u,v] \leftarrow S_{\text{shr}}[u,v] + \Delta_t, \qquad \forall (u,v) \in \mathcal{P}_t \text{ such that } \ell_t \in \mathcal{L}(v)$$

  else ▷ **Unshared branch**
    Back-propagation: Propagate multiplicative updates to S_{unshr} along sampled risky ancestors:

$$S_{\text{unshr}}[u,v] \leftarrow S_{\text{unshr}}[u,v] \cdot \exp\left(-\eta \frac{c_t}{\Pi_t(u)\, p_t(v \mid u)}\right), \qquad \forall (u,v) \in \mathcal{P}_t \text{ with } u \text{ risky}$$

  end if
end for

Algorithm 2 Adaptive Partial-Share (PS) Algorithm

Input: Tree $\mathcal{T}$, Shared leaves $\mathcal{L}_{\text{shr}}$, learning rate η, exploration rate ϵ.
Pre-processing: Label node i as *Safe* if subtree$(i) \subseteq \mathcal{L}_{\text{shr}}$, else *Risky*.
Initialize: $\theta_\ell \leftarrow 0, \forall \ell \in \mathcal{L}$.
Compute initial aggregates for all edges (i, j):

$$S_{shr}[i,j] \leftarrow \sum_{\ell \in \text{subtree}(j) \cap \mathcal{L}_{\text{shr}}} \exp(\eta\theta_\ell), \quad S_{unshr}[i,j] \leftarrow \exp\Big(\sum_{\ell \in \text{subtree}(j) \cap \mathcal{L}_{\text{unshr}}} \eta\theta_\ell \Big)$$

for each round $t = 1, \ldots, T$ **do**
  1. Adaptive Selection:
  $i \leftarrow root, \Pi_t(i) \leftarrow 1$
  while i is not a leaf **do**
    Calculate exploitation probability based on subtree exponential weights:

$$P_{exp}(j) = \frac{S_{shr}[i,j] + S_{unshr}[i,j]}{\sum_{k \in Children(i)} (S_{shr}[i,k] + S_{unshr}[i,k])}$$

    Determine local exploration rate: $\epsilon_i \leftarrow \epsilon$ if i is *Risky* else 0.
    Construct sampling distribution for child j:

$$p_t(j) = (1 - \epsilon_i) P_{exp}(j) + \frac{\epsilon_i}{|Children(i)|}$$

    Sample child $j \sim p_t(\cdot)$.
    Update path probability for child j:

$$\Pi_t(j) \leftarrow \Pi_t(i) \cdot p_t(j)$$

    $i \leftarrow j$
  end while
  Let leaf $\ell_t \leftarrow i$. Observe cost c_t.
  2. Hybrid Update:
  $val_{old} \leftarrow \exp(\eta\theta_{\ell_t})$ ▷ Record leaf weight before update
  if $\ell_t \in \mathcal{L}_{\text{shr}}$ **then** ▷ **Full-Information Mode**
    $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - c_t$
    Calculate change: $\Delta \leftarrow \exp(\eta\theta_{\ell_t}) - val_{old}$
    Back-propagation: Propagate Δ to S_{shr} along ancestors:

$$S_{shr}[parent, child] \leftarrow S_{shr}[parent, child] + \Delta$$

  else ▷ **Bandit Mode**
    $\theta_{old} \leftarrow \theta_{\ell_t}$ ▷ (modified) record θ before update
    $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - c_t / \Pi_t(\ell_t)$
    (modified) Compute multiplicative factor:

$$r \leftarrow \exp\big(\eta(\theta_{\ell_t} - \theta_{old})\big)$$

    (modified) Back-propagation: Update S_{unshr} along ancestors multiplicatively:

$$S_{unshr}[parent, child] \leftarrow S_{unshr}[parent, child] \cdot r$$

  end if
end for

Algorithm 3 Full-Share Algorithm (Recursive Aggregation)

Input: Tree $\mathcal{T}$, learning rate η
Initialize:
  $\theta_\ell \leftarrow 0, \forall \ell \in \mathcal{L}$.
  $W[u] \leftarrow \sum_{\ell \in \text{subtree}(u)} \exp(\eta \cdot \theta_\ell), \forall u \in \mathcal{T}$.
for each round $t = 1, \ldots, T$ **do**
  Selection:
  $i \leftarrow root$
  while i is not a leaf **do**
    Select child j from $Children(i)$ with probability:

$$P(j|i) = \frac{\sum_{\ell \in \text{subtree}(j)} \exp(\eta \cdot \theta_\ell)}{\sum_{\ell \in \text{subtree}(i)} \exp(\eta \cdot \theta_\ell)} = \frac{W[j]}{W[i]}$$

    $i \leftarrow j$
  end while
  Let leaf $\ell_t \leftarrow i$.
  Observe cost c_t at leaf ℓ_t.
  Update:
  $\theta_{old} \leftarrow \theta_{\ell_t}$
  $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - \frac{c_t}{\Pi_t(\ell_t)}$
  Propagate Change:
  Calculate weight change at leaf:

$$\Delta = \exp(\eta \cdot \theta_{\ell_t}) - \exp(\eta \cdot \theta_{old})$$

  $curr \leftarrow \ell_t$
  while $curr \neq root$ **do**
    $W[curr] \leftarrow W[curr] + \Delta$
    $curr \leftarrow Parent(curr)$
  end while
  $W[root] \leftarrow W[root] + \Delta$
end for

Definitions

For any **risky** non-leaf node i, let C_i be the set of its children with $d_i = |C_i|$. Let $\Pi_t(i)$ denote the probability that node i is reached in round t. For any child $j \in C_i$, we define $\rho_{j,t}$ as the conditional probability that a leaf selected from the subtree of j is an **unshared** leaf:

$$\rho_{j,t} := \mathbb{P}(\text{leaf} \in \mathcal{L}_{\text{unshr}} \mid J_t = j).$$

We define the **Effective Unshared Mass** of node i at round t as

$$\Lambda_t(i) = \sum_{j \in C_i} \rho_{j,t}. \tag{1}$$

We also define the **local exploration rate** at node i as

$$\epsilon_i := \begin{cases} 0, & \text{if } C_i \subseteq \mathcal{L}, \\ \epsilon, & \text{otherwise.} \end{cases} \tag{2}$$

Lemma PS-2: Moments of the Estimator at a Risky Node

Lemma 1 (Exact Second Moment Bound at a Risky Node)**.** *For any **risky** non-leaf node i, let $z[j,t]$ be the estimated cost for child j constructed by the PS algorithm. Let $\mathcal{H}_{t-1}$ be the history up to round $t-1$. Then the conditional weighted second moment of the estimators satisfies:*

$$\sum_{j \in C_i} p_t(j \mid i)\, \mathbb{E}_t\big[z[j,t]^2\big] \le \underbrace{\sum_{j \in C_i} p_t(j \mid i)\,(1-\rho_{j,t})}_{\textit{Shared Mass}} + \underbrace{\frac{1}{\Pi_t(i)^2} \sum_{j \in C_i} \rho_{j,t}}_{\textit{Unshared Variance}}. \tag{3}$$

Lemma PS-3: Equivalence of Virtual and Effective Expected Costs

Lemma 2 (Expectation Bridge)**.** *Let $\tilde{y}[k,t] \triangleq \mathbb{E}_t[z[k,t] \mid \mathcal{H}_{t-1}]$ be the **expected estimator value** (effective cost) for child k. Then the expected virtual loss (used by Hedge) satisfies:*

$$\mathbb{E}_t\left[\sum_{j \in C_i} p_t(j|i) z[j,t] \middle| \mathcal{H}_{t-1}\right] = \mathbb{E}_{k \sim p_t(\cdot|i)}\left[\tilde{y}[k,t]\right]. \tag{4}$$

Theorem PS-1: Single-Layer Regret Bound at a Risky Node

Theorem 1 (Single-Layer Regret of PS at a Risky Node)**.** *For any **risky** non-leaf node i, the cumulative regret with respect to the optimal child j^* satisfies*

$$\sum_{t=1}^{T} \mathbb{E}[c_t(i)] - \sum_{t=1}^{T} c_t(j^*) \le \epsilon_i T + \frac{\ln d_i}{\eta} + \eta \sum_{t=1}^{T} \left(\Pi_t(i) \sum_{k \in C_i} p_t(k|i)(1-\rho_{k,t}) + \frac{\Lambda_t(i)}{\Pi_t(i)} \right). \tag{5}$$

A Proofs

A.1 Proof of Lemma 1

Proof. We analyze the conditional second moment $\mathbb{E}_t[z[j,t]^2]$ according to whether the selected leaf inside the subtree of child j is shared or unshared. Recall that

$$\rho_{j,t} = \mathbb{P}(\text{leaf} \in \mathcal{L}_{\text{unshr}} \mid J_t = j).$$

For a fixed child $j \in C_i$,

$$\begin{aligned}
\mathbb{E}_t[z[j,t]^2] &= (1-\rho_{j,t}) \cdot \mathbb{E}_t[z[j,t]^2 \mid \text{Shared}] + \rho_{j,t} \cdot \mathbb{E}_t[z[j,t]^2 \mid \text{Unshared}] \\
&= (1-\rho_{j,t}) \cdot y[j,t]^2 + \rho_{j,t} \cdot \mathbb{E}_t\left[\left(\frac{y[j,t]\mathbb{I}(J_t=j)}{p_t(j|i)\Pi_t(i)}\right)^2 \middle| \text{Unshared}\right] \\
&\leq (1-\rho_{j,t}) \cdot 1 + \rho_{j,t} \cdot \frac{1}{(p_t(j|i)\Pi_t(i))^2}\mathbb{E}_t[\mathbb{I}(J_t=j)] \\
&= (1-\rho_{j,t}) + \frac{\rho_{j,t}}{p_t(j|i)\Pi_t(i)^2},
\end{aligned}$$

where we used $y[j,t] \leq 1$ and $\mathbb{E}_t[\mathbb{I}(J_t=j)] = p_t(j|i)$.

Multiplying both sides by $p_t(j|i)$ and summing over $j \in C_i$ gives

$$\sum_{j\in C_i} p_t(j|i)\mathbb{E}_t[z[j,t]^2] \leq \sum_{j\in C_i} p_t(j|i)(1-\rho_{j,t}) + \frac{1}{\Pi_t(i)^2}\sum_{j\in C_i}\rho_{j,t}.$$

This proves the claim. □

A.2 Proof of Lemma 2

Proof. By linearity of expectation,

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{j\in C_i} p_t(j|i)z[j,t]\right] &= \sum_{j\in C_i} p_t(j|i)\mathbb{E}_t[z[j,t]] \\
&= \sum_{j\in C_i} p_t(j|i)\tilde{y}[j,t].
\end{aligned}$$

By definition, the right-hand side is exactly $\mathbb{E}_{k\sim p_t(\cdot|i)}[\tilde{y}[k,t]]$.

To make $\tilde{y}[k,t]$ explicit, distinguish two cases:

- **Case 1: Shared subtree under child k.** Then the estimator is deterministic:

$$\tilde{y}[k,t] = \mathbb{E}_t[z[k,t]] = y[k,t].$$

- **Case 2: Unshared subtree under child k.** The estimator is IPS:

$$z[k,t] = \frac{y[k,t]}{p_t(k|i)\Pi_t(i)}\mathbb{I}(J_t=k).$$

  Hence

$$\tilde{y}[k,t] = \mathbb{E}_t[z[k,t]] = p_t(k|i) \cdot \frac{y[k,t]}{p_t(k|i)\Pi_t(i)} = \frac{y[k,t]}{\Pi_t(i)}.$$

Therefore the bridge identity holds. □

A.3 Proof of Theorem 1

Proof. Apply the standard Hedge regret bound to the estimator sequence $\{z[j,t]\}_{j\in C_i, t\le T}$:

$$\sum_{t=1}^{T}\sum_{j\in C_i} p_t(j|i)z[j,t] \le \sum_{t=1}^{T} z[j^*,t] + \frac{\ln d_i}{\eta} + \eta\sum_{t=1}^{T}\sum_{j\in C_i} p_t(j|i)z[j,t]^2. \tag{6}$$

Taking total expectation over the event that node i is reached, Lemma 1 gives

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T}\sum_{j\in C_i} p_t(j|i)z[j,t]^2\right] &= \sum_{t=1}^{T}\Pi_t(i)\cdot\mathbb{E}_t\left[\sum_{j\in C_i} p_t(j|i)z[j,t]^2\right] \\
&\le \sum_{t=1}^{T}\Pi_t(i)\left(\sum_{j\in C_i} p_t(j|i)(1-\rho_{j,t}) + \frac{\Lambda_t(i)}{\Pi_t(i)^2}\right) \\
&= \sum_{t=1}^{T}\left(\Pi_t(i)\sum_{j\in C_i} p_t(j|i)(1-\rho_{j,t}) + \frac{\Lambda_t(i)}{\Pi_t(i)}\right).
\end{aligned}$$

Combining the above inequality with Lemma 2, and noting that the exploration mixture contributes an additive $\epsilon_i T$ term, we obtain

$$\sum_{t=1}^{T}\mathbb{E}[c_t(i)] - \sum_{t=1}^{T} c_t(j^*) \le \epsilon_i T + \frac{\ln d_i}{\eta} + \eta\sum_{t=1}^{T}\left(\Pi_t(i)\sum_{k\in C_i} p_t(k|i)(1-\rho_{k,t}) + \frac{\Lambda_t(i)}{\Pi_t(i)}\right),$$

which proves (5). □

B Recursive Root Regret Bound for Risky-PS

Definitions

For any node i, define the cumulative expected loss of the algorithm starting from i by

$$Y_\eta[i,T] \;\triangleq\; \sum_{t=1}^{T} \mathbb{E}[\, y_\eta[i,t]\,].$$

Let $Y_*[i,T]$ denote the cumulative loss of the optimal stationary policy starting from node i, defined recursively as

$$Y_*[i,T] \;=\; \begin{cases} \sum_{t=1}^{T} c_t(i), & \text{if } i \in \mathcal{L}, \\ \min_{j\in C_i} Y_*[j,T], & \text{otherwise.} \end{cases}$$

For any non-leaf node i, define its local branching factor and optimal-child set by

$$d_i \;\triangleq\; |C_i|, \qquad C_i^* \;\triangleq\; \arg\min_{j\in C_i} Y_*[j,T].$$

We further define the *dangerous-child count* of node i as

$$D^U(i) \;\triangleq\; |\{j \in C_i :\ \text{subtree}(j) \cap \mathcal{L}_{\text{unshr}} \neq \emptyset\}|\,.$$

To make the risky depth precise, define the *risky height* of each node i recursively by

$$R(i) \;\triangleq\; \begin{cases} 0, & \text{if } i \text{ is safe,} \\ 1, & \text{if } i \text{ is risky and } C_i \subseteq \mathcal{L}, \\ 1 + \max_{j\in C_i} R(j), & \text{if } i \text{ is risky and } C_i \not\subseteq \mathcal{L}. \end{cases}$$

We then define the *risky skeleton depth* of the whole tree as

$$R \;\triangleq\; R(r),$$

where r is the root. Equivalently, R is the maximum number of risky nodes that can appear on any root-to-leaf path before the path enters a fully safe suffix.

For risky nodes, we use the R-adjusted parameters

$$\eta_i^{\text{risk}} \;=\; T^{-\frac{R}{R+1}}, \qquad \epsilon_i \;=\; \begin{cases} 0, & C_i \subseteq \mathcal{L}, \\ d_i\, T^{-\frac{1}{R+1}}, & C_i \not\subseteq \mathcal{L}. \end{cases}$$

For a safe node i, let

$$N_i \;\triangleq\; |\mathcal{L}(i)|$$

be the number of leaves in the safe subtree rooted at i. We upper bound its local contribution by the standard N_i-armed Exp3 rate over at most T rounds, using

$$\eta_i^{\text{shr}} \;=\; \sqrt{\frac{\ln N_i}{T\, N_i}}.$$

Lemma RR-1 (Safe-Subtree Exp3 Equivalence). For any safe node i, let

$$N_i \;\triangleq\; |\mathcal{L}(i)|$$

denote the number of leaves in the safe subtree rooted at i. Then the recursive selection rule of the full-share algorithm restricted to the subtree of i is equivalent to running an N_i-armed Exp3 forecaster directly on the leaf set $\mathcal{L}(i)$.

More precisely, if the leaf weights are

$$w_t(\ell) \;\triangleq\; \exp\!\big(\eta_i^{\text{shr}} \theta_{\ell,t}\big), \qquad \ell \in \mathcal{L}(i),$$

and the recursive aggregate at any node u is

$$W_t[u] \triangleq \sum_{\ell \in \mathrm{subtree}(u)} w_t(\ell),$$

then for any leaf $\ell \in \mathcal{L}(i)$, the probability that the recursive aggregation procedure starting from node i selects ℓ at round t is exactly

$$P_t(\ell \mid i) = \frac{w_t(\ell)}{\sum_{\ell' \in \mathcal{L}(i)} w_t(\ell')} = \frac{\exp(\eta_i^{\mathrm{shr}} \theta_{\ell,t})}{\sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\mathrm{shr}} \theta_{\ell',t})}. \tag{7}$$

Consequently, viewing the safe subtree as an N_i-armed Exp3 problem over at most T rounds, we obtain

$$Y_\eta[i,T] \;\le\; Y_*[i,T] \;+\; 2\sqrt{T\, N_i \ln N_i}. \tag{8}$$

Proof. Since node i is safe, every leaf in subtree(i) is shared. The recursive aggregation algorithm selects a child j of an internal node u with probability

$$P_t(j \mid u) = \frac{W_t[j]}{W_t[u]}.$$

Fix any leaf $\ell \in \mathcal{L}(i)$, and let

$$i = u_0 \to u_1 \to \cdots \to u_m = \ell$$

be the unique path from node i to leaf ℓ. The probability that the recursive procedure starting at i finally reaches ℓ is the product of the transition probabilities along this path:

$$P_t(\ell \mid i) = \prod_{r=0}^{m-1} P_t(u_{r+1} \mid u_r) = \prod_{r=0}^{m-1} \frac{W_t[u_{r+1}]}{W_t[u_r]}.$$

This product telescopes:

$$P_t(\ell \mid i) = \frac{W_t[\ell]}{W_t[i]}.$$

Since a leaf has no descendants except itself,

$$W_t[\ell] = w_t(\ell) = \exp(\eta_i^{\mathrm{shr}} \theta_{\ell,t}),$$

and since i is the root of the current safe subtree,

$$W_t[i] = \sum_{\ell' \in \mathcal{L}(i)} w_t(\ell') = \sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\mathrm{shr}} \theta_{\ell',t}).$$

Therefore,

$$P_t(\ell \mid i) = \frac{\exp(\eta_i^{\mathrm{shr}} \theta_{\ell,t})}{\sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\mathrm{shr}} \theta_{\ell',t})},$$

which is exactly the arm-selection probability of an exponential-weights/Exp3 forecaster run directly on the leaf set $\mathcal{L}(i)$.

Hence, the safe subtree rooted at i can be viewed as an N_i-armed adversarial bandit problem. Since node i can be visited in at most T rounds, a valid upper bound is obtained by applying the standard Exp3 regret bound over horizon T with learning rate

$$\eta_i^{\mathrm{shr}} = \sqrt{\frac{\ln N_i}{T N_i}}.$$

This gives

$$Y_\eta[i,T] - Y_*[i,T] \;\le\; 2\sqrt{T\, N_i \ln N_i},$$

which proves the claim. □

Lemma RR-2 (R-Adjusted Local Risky Bounds). For any risky node i, the single-layer regret satisfies the following fixed-parameter bounds:

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}}, \qquad \text{if } C_i \subseteq \mathcal{L}, \tag{8}$$

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(d_i + D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}}, \qquad \text{if } C_i \not\subseteq \mathcal{L}. \tag{9}$$

Proof. This follows from the single-layer Partial-Share bound (Theorem PS-1) after substituting the R-adjusted risky parameters

$$\eta_i^{\text{risk}} = T^{-\frac{R}{R+1}}, \qquad \epsilon_i = \begin{cases} 0, & C_i \subseteq \mathcal{L}, \\ d_i\, T^{-\frac{1}{R+1}}, & C_i \not\subseteq \mathcal{L}, \end{cases}$$

together with $|C_i| = d_i$ and the structural bound $\Lambda_t(i) \le D^U(i)$. When $C_i \subseteq \mathcal{L}$, the exploration term vanishes, yielding (8). Otherwise, the extra exploration contribution produces the additional $d_i\, T^{\frac{R}{R+1}}$ term, yielding (9). □

Theorem RR-1 (Recursive Root Regret Bound of Risky-PS). Define recursively, for every node i,

$$\Delta(i) \;\triangleq\; \begin{cases} 0, & i \in \mathcal{L}, \\ 2\sqrt{T\, N_i \ln N_i}, & i \text{ is safe}, \\ \big(D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}}, & i \text{ is risky and } C_i \subseteq \mathcal{L}, \\ \big(d_i + D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}} + \min\limits_{j\in C_i^*} \Delta(j), & i \text{ is risky and } C_i \not\subseteq \mathcal{L}. \end{cases} \tag{10}$$

Then for every node i,

$$Y_\eta[i,T] \;\le\; Y_*[i,T] + \Delta(i). \tag{11}$$

In particular, for the root node r,

$$\mathrm{Reg}_{\text{root}}(T) \;\triangleq\; Y_\eta[r,T] - Y_*[r,T] \;\le\; \Delta(r). \tag{12}$$

Proof. We proceed by structural induction on the subtree rooted at i.

Base case: If $i \in \mathcal{L}$, then by definition

$$Y_\eta[i,T] = Y_*[i,T],$$

and since $\Delta(i) = 0$, (11) holds.

Safe case: If i is safe, then (11) follows directly from Lemma RR-1:

$$Y_\eta[i,T] \le Y_*[i,T] + 2\sqrt{T\, N_i \ln N_i} = Y_*[i,T] + \Delta(i).$$

Risky leaf-parent case: If i is risky and $C_i \subseteq \mathcal{L}$, then Lemma RR-2 gives

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}}.$$

Since all children are leaves,

$$\min_{j\in C_i} Y_\eta[j,T] = \min_{j\in C_i} Y_*[j,T] = Y_*[i,T].$$

Therefore,

$$Y_\eta[i,T] \le Y_*[i,T] + \big(D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}} = Y_*[i,T] + \Delta(i).$$

Risky non-leaf case: Suppose now that i is risky and $C_i \not\subseteq \mathcal{L}$. By Lemma RR-2,

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(d_i + D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}}. \tag{13}$$

Take any optimal child $j^* \in C_i^*$. By definition of C_i^*,

$$Y_*[j^*,T] = Y_*[i,T]. \tag{14}$$

By the induction hypothesis applied to node j^*,

$$Y_\eta[j^*, T] \le Y_*[j^*, T] + \Delta(j^*). \tag{15}$$

Also,

$$\min_{j \in C_i} Y_\eta[j, T] \le Y_\eta[j^*, T]. \tag{16}$$

Substituting (15) and (16) into (13), and then using (14), yields

$$Y_\eta[i, T] \le Y_*[i, T] + \Delta(j^*) + \big(d_i + D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}}.$$

Since this holds for any $j^* \in C_i^*$, taking the minimum over C_i^* gives

$$Y_\eta[i, T] \le Y_*[i, T] + \min_{j \in C_i^*} \Delta(j) + \big(d_i + D^U(i) + \ln d_i\big)\, T^{\frac{R}{R+1}} = Y_*[i, T] + \Delta(i).$$

Thus the induction closes, and (11) holds for all nodes i. Applying it to the root proves (12). □

Sanity checks.

- **All-share.** If the whole tree is safe, then r is safe and

  $$\Delta(r) = 2\sqrt{T\,|\mathcal{L}|\,\ln|\mathcal{L}|},$$

  which is independent of the depth L.

- **All-unshare.** If all leaves are unshared, then every internal node is risky and $D^U(i) = d_i$ for every internal node i. Hence the recursive expansion yields

  $$\Delta(r) = \left(\sum_{i \in \mathcal{P}_{\text{int}}(r)} d_i + \sum_{i \in \mathcal{P}_{\text{int}}(r) \setminus \{i_{\text{leaf-parent}}\}} d_i + \sum_{i \in \mathcal{P}_{\text{int}}(r)} \ln d_i \right) T^{\frac{L}{L+1}},$$

  where $\mathcal{P}_{\text{int}}(r)$ denotes the set of internal nodes on the optimal root-to-leaf path, and $i_{\text{leaf-parent}}$ is its last internal node. In the homogeneous case $d_i \equiv D$, this simplifies to

  $$\Delta(r) = \big((2L-1)D + L\ln D\big)\, T^{\frac{L}{L+1}},$$

  which exactly recovers the original worst-case endpoint.

C A Structure-Dependent Lower Bound for Risky Partial-Share

In this section, we establish a regret lower bound for a class of time-homogeneous oracle policies under a tree whose longest risky prefix has length R. This lower bound shows that even when all side branches are fully shared safe subtrees, the need to *educate* a deepest risky learner still induces a polynomial regret barrier.

Risky-prefix system. We consider a tree constructed as follows. There are R risky internal nodes, denoted by $v_1, v_2, \ldots, v_R$, where v_1 is the root. For each $i \le R-1$, node v_i has two children: (i) a fully-shared safe subtree $\mathcal{S}_i$, and (ii) the next risky node v_{i+1}. The node v_R has two leaf children u and u', both of which are unshared. By construction, once a path enters any $\mathcal{S}_i$, all remaining descendants are shared and hence safe; therefore the longest risky prefix is exactly R.

Each visited leaf incurs a Bernoulli cost. Fix a small positive constant $\delta < 1/2^R$. Let

$$p^* \triangleq \frac{1 - 2^R\delta}{2}.$$

We set one of $p_u, p_{u'}$ to p^* and the other to

$$\frac{1 + 2^R\delta}{2}.$$

For each safe subtree $\mathcal{S}_i$, all of its leaves are shared and have the same Bernoulli mean

$$p_{\mathcal{S}_i} \triangleq \frac{1-(2^R-2^{i-1})\delta}{2}, \qquad i=1,2,\dots,R-1.$$

Hence, for any t,

$$\mathbb{E}[y[\mathcal{S}_i,t]] = p_{\mathcal{S}_i},$$

regardless of which shared leaf inside $\mathcal{S}_i$ is reached. In particular,

$$p_{\mathcal{S}_1} - p^* = \frac{\delta}{2}.$$

The regret of the system is

$$\sum_{t=1}^{T} \mathbb{E}[y[v_1,t]] - p^*T = \sum_{t=1}^{T}\left(\mathbb{E}[y[v_1,t]] - \frac{1-2^R\delta}{2}\right). \tag{7}$$

We consider the following policy class: the deepest risky node v_R may employ an arbitrary bandit learning algorithm with feedback only when it receives a job, whereas nodes $v_1,\dots,v_{R-1}$ employ time-homogeneous oracle policies in the sense of Definition 2. That is, before deciding between $\mathcal{S}_i$ and v_{i+1}, node v_i knows the expected costs of its two children, and if the absolute gap equals ζ, it forwards the job to the higher-cost child with probability $P_i(\zeta)$ and to the other child with probability $1-P_i(\zeta)$, where $P_i(\cdot)$ is an arbitrary decreasing function.

We note that the fully-shared structure of each $\mathcal{S}_i$ is precisely where the *share* assumption enters the lower bound: the side branch is safe, and its expected cost is completely revealed to its parent. Thus, the only nontrivial learning challenge lies on the risky prefix $v_1 \to v_2 \to \cdots \to v_R$.

Lemma 3. *There exists a positive integer N_δ such that, for all t with $N_R(t) < N_\delta$,*

$$\mathbb{E}[y[v_R,t]] > \frac{1-(2^R-2^{R-1})\delta}{2},$$

where $N_R(t)$ denotes the number of jobs that node v_R has received from its parent up to time t.

Proof. This is the same bottom-level two-armed bandit hardness argument as in Lemma 5 of the original proof, with L replaced by R. Since node v_R can only learn the means of u and u' from the rounds in which it actually receives a job, there exists a finite threshold N_δ such that before receiving N_δ samples, its expected cost cannot be reduced below $\big(1-(2^R-2^{R-1})\delta\big)/2$. □

Theorem 2. *For the risky-prefix system above, the regret is*

$$\Omega\Big(T^{\frac{R-1}{R}}\Big)$$

for any bandit learning policy employed by node v_R and any time-homogeneous oracle policies employed by nodes $v_1,\dots,v_{R-1}$. Consequently, the same lower bound applies a fortiori *to Risky Partial-Share, which has no more information than this oracle class.*

Proof. Let T_δ be the time at which $N_R(t) = N_\delta$. By Lemma 3,

$$\mathbb{E}[y[v_R,t]] > \frac{1-(2^R-2^{R-1})\delta}{2} \qquad \text{for all } t < T_\delta.$$

We first study the system behavior before time T_δ. Consider the forwarding decision of node v_{R-1} at any time $t < T_\delta$. Node v_{R-1} has two children: one is the shared safe subtree $\mathcal{S}_{R-1}$ with

$$\mathbb{E}[y[\mathcal{S}_{R-1},t]] = p_{\mathcal{S}_{R-1}} = \frac{1-(2^R-2^{R-2})\delta}{2},$$

and the other is the non-leaf node v_R with

$$\mathbb{E}[y[v_R,t]] > \frac{1-(2^R-2^{R-1})\delta}{2} = p_{\mathcal{S}_{R-1}} + 2^{R-3}\delta.$$

By Assumption A2, the probability that node v_{R-1} selects the higher-cost child v_R is at most

$$q_{R-1} \triangleq P_{R-1}(2^{R-3}\delta).$$

We also have

$$\mathbb{E}[y[v_{R-1}, t]] \geq p_{\mathcal{S}_{R-1}} = \frac{1 - (2^R - 2^{R-2})\delta}{2}.$$

We further analyze the forwarding decision of node v_i for $i < R-1$ at any time $t < T_\delta$. Using a simple induction argument, it can be shown that the probability that node v_i selects node v_{i+1} is at most

$$q_i \triangleq P_i(2^{i-2}\delta),$$

and

$$\mathbb{E}[y[v_i, t]] \geq p_{\mathcal{S}_i} = \frac{1 - (2^R - 2^{i-1})\delta}{2}.$$

Therefore, at any time $t < T_\delta$, we have

$$\mathbb{E}[y[v_1, t]] - \frac{1 - 2^R\delta}{2} \geq \frac{\delta}{2}. \tag{8}$$

Moreover, since node v_R can only receive a job if, for each $i \leq R-1$, node v_i selects node v_{i+1}, which happens with probability at most q_i, we have

$$\mathbb{E}[T_\delta] \geq \frac{N_\delta}{\prod_{i=1}^{R-1} q_i}. \tag{9}$$

Next, we analyze the system behavior after time T_δ. For any time $t > T_\delta$,

$$\mathbb{E}[y[v_R, t]] \geq \min\{p_u, p_{u'}\} = \frac{1 - 2^R\delta}{2}.$$

Consider the forwarding decision of node v_{R-1}. Since

$$\mathbb{E}[y[\mathcal{S}_{R-1}, t]] = p_{\mathcal{S}_{R-1}} = \frac{1 - (2^R - 2^{R-2})\delta}{2} \leq \mathbb{E}[y[v_R, t]] + 2^{R-3}\delta,$$

the probability that node v_{R-1} selects node v_R is at most

$$1 - P_{R-1}(2^{R-3}\delta) = 1 - q_{R-1}.$$

Using a simple induction argument, we can further show that the probability that node v_i selects node v_{i+1} is at most $1 - q_i$, for all $i \leq R-1$. Hence, the probability that node v_R receives a job is at most

$$\prod_{i=1}^{R-1}(1 - q_i).$$

If node v_R does not receive a job, which happens with probability at least

$$1 - \prod_{i=1}^{R-1}(1 - q_i),$$

then the job must enter one of the safe subtrees $\mathcal{S}_1, \ldots, \mathcal{S}_{R-1}$, and the expected cost is at least

$$\min_{1 \leq i \leq R-1} p_{\mathcal{S}_i} = p_{\mathcal{S}_1} = \frac{1 - (2^R - 1)\delta}{2}.$$

We then have, at any time $t > T_\delta$,

$$\mathbb{E}[y[v_1, t]] - \frac{1 - 2^R\delta}{2} \geq \left(1 - \prod_{i=1}^{R-1}(1 - q_i)\right)\frac{\delta}{2}. \tag{10}$$

Combining (8), (9), and (10), we obtain the following regret bound:

$$\sum_{t=1}^{T}\left(\mathbb{E}[y[v_1,t]]-\frac{1-2^R\delta}{2}\right)\geq\mathbb{E}\left[\sum_{t=1}^{T_\delta}\frac{\delta}{2}+\sum_{t=T_\delta+1}^{T}\left(1-\prod_{i=1}^{R-1}(1-q_i)\right)\frac{\delta}{2}\right]$$
$$\geq\frac{\delta}{2}\left[\frac{N_\delta}{\prod_{i=1}^{R-1}q_i}+\left(T-\frac{N_\delta}{\prod_{i=1}^{R-1}q_i}\right)\left(1-\prod_{i=1}^{R-1}(1-q_i)\right)\right]. \tag{11}$$

It is then straightforward to show that

$$\frac{\delta}{2}\left[\frac{N_\delta}{\prod_{i=1}^{R-1}q_i}+\left(T-\frac{N_\delta}{\prod_{i=1}^{R-1}q_i}\right)\left(1-\prod_{i=1}^{R-1}(1-q_i)\right)\right]=\Omega\left(T^{\frac{R-1}{R}}\right).$$

Moreover, setting

$$q_i=\Theta\left(T^{-1/R}\right),\qquad\forall\, i\leq R-1,$$

makes

$$\frac{\delta}{2}\left[\frac{N_\delta}{\prod_{i=1}^{R-1}q_i}+\left(T-\frac{N_\delta}{\prod_{i=1}^{R-1}q_i}\right)\left(1-\prod_{i=1}^{R-1}(1-q_i)\right)\right]=\Theta\left(T^{\frac{R-1}{R}}\right).$$

This completes the proof. □

D All-Share Lower Bound via Adversarial Bandit Reduction

Assume the whole tree is all-share, so that the root r is safe. Let

$$N\ \triangleq\ |\mathcal{L}(r)|$$

be the number of leaves. Then the recursive full-share selection rule induces an N-armed adversarial bandit policy on the leaf set $\mathcal{L}(r)$. Consequently, there exists a universal constant $c_0>0$ such that for any $\delta\in(0,1)$ satisfying the conditions of Theorem 17.4, there exists a cost sequence $\{c_t(\ell)\}_{t\leq T,\ell\in\mathcal{L}(r)}\subset[0,1]$ such that

$$\mathbb{P}\left(\mathrm{Reg}_{\mathrm{root}}(T)\ \geq\ c_0\sqrt{T\,N\,\log\frac{1}{2\delta}}\right)\ \geq\ \delta. \tag{LB-1}$$

In particular, there exists a universal constant $c_1>0$ such that

$$\sup_{c_{1:T}}\mathbb{E}[\mathrm{Reg}_{\mathrm{root}}(T)]\ \geq\ c_1\sqrt{T\,N}. \tag{LB-2}$$

Proof. For any leaf $\ell\in\mathcal{L}(r)$, let

$$r=u_0\to u_1\to\cdots\to u_m=\ell$$

be the unique root-to-leaf path. Under the recursive aggregation rule,

$$P_t(u_{s+1}\mid u_s)=\frac{W_t[u_{s+1}]}{W_t[u_s]},$$

hence the probability of selecting leaf ℓ at round t is

$$P_t(\ell\mid r)=\prod_{s=0}^{m-1}\frac{W_t[u_{s+1}]}{W_t[u_s]}=\frac{W_t[\ell]}{W_t[r]}=\frac{w_t(\ell)}{\sum_{\ell'\in\mathcal{L}(r)}w_t(\ell')}.$$

Therefore, the recursive full-share algorithm on the all-share tree is equivalent, at the leaf level, to an N-armed bandit policy over $\mathcal{L}(r)$.

Now define rewards by

$$x_{t\ell}\ \triangleq\ 1-c_t(\ell),\qquad\ell\in\mathcal{L}(r).$$

Then the reward regret and the cost regret coincide:

$$\max_{\ell\in\mathcal{L}(r)}\sum_{t=1}^{T}(x_{t\ell}-x_{tA_t})=\sum_{t=1}^{T}c_t(A_t)-\min_{\ell\in\mathcal{L}(r)}\sum_{t=1}^{T}c_t(\ell)=\mathrm{Reg}_{\mathrm{root}}(T).$$

Applying Theorem 17.4 with $k=N$ and $n=T$ yields the high-probability lower bound (LB-1). Choosing a fixed constant δ (e.g. $\delta=1/4$) gives (LB-2). □